



Dual-domain solute transfer and transport processes: evaluation in batch and transport experiments

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Received 10 February 2004; received in revised form 28 June 2004; accepted 9 July 2004

Abstract

Soil macropore networks establish a dual-domain transport scenario in which water and solutes are preferentially channeled through soil macropores while slowly diffusing into and out of the bulk soil matrix. The influence of macropore networks on intra-ped solute diffusion and preferential transport in a soil typical of subsurface-drained croplands in the Midwestern United States was studied in batch- and column-scale experiments. In the batch diffusion studies with soil aggregates, the estimated diffusion radius (length) of the soil aggregates corresponded to the half-spacing of the aggregate fissures, suggesting that the intra-ped fissures reduced the diffusion impedance and preferentially allowed solutes to diffuse into the soil matrix. In the column-scale solute transport experiments, the average diffusion radius (estimated from HYDRUS-2D simulations and a first-order diffusive transfer term) was nearly double that of the batch-scale study. This increase may be attributed to a loss of pore continuity and a compounding of the small diffusion impedance through macropores at the larger scale. The column-scale solute transport experiments also suggest that two preferential networks exist in the soil. At and near soil saturation, a primary network of large macropores (possibly root channels and earthworm burrows) dominate advective transport, causing a high degree of physical and sorption nonequilibrium and simultaneous breakthrough of a nonreactive (bromide) and a reactive (alachlor) solute. As the saturation level decreases, the primary network drains, while transport through smaller macropores (possibly intra-ped features) continues, resulting

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in a reduced degree of nonequilibrium and separation in the breakthrough curves of bromide andalachlor.

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Keywords: Preferential transport; Nonequilibrium; Dual-porosity; Macropore flow; Mass transfer; Diffusion

1. Introduction

Describing water and solute movement in agricultural field soils is often complicated by the presence of macropore networks formed from root channels, earthworm burrows, and intra-pedal voids (Beven and Germann, 1982; Willoughby and Kladvko, 2002). Preferential transport of water and dissolved constituents through these macropores results in limited interaction with the bulk soil matrix via intra-ped water and mass exchange, leading to rapid solute breakthrough (Johnson et al., 2003), extended tailing (Rao et al., 1980b; Brusseau and Rao, 1990; van Genuchten and Wierenga, 1976), and long leaching periods (Rao et al., 1974)—even for nonreactive solutes (Balasubramanian et al., 1973). The mathematics of intra-particle solute diffusion are well described using Fick's first and second laws (cf., Crank, 1975); however, the quantification of intra-ped diffusion is difficult due to its dependence on pore tortuosity (Buckingham, 1904; Millington and Quirk, 1959; Millington and Shearer, 1971), structural geometry (Rao et al., 1980a; van Genuchten and Dalton, 1986; Valocchi, 1990), pore-size distribution (Moldrup et al., 1999; Olesen et al., 2001a,b; Moldrup et al., 2003), and degree of water saturation (Moldrup et al., 1996; Olesen et al., 2001a). Recent studies highlight the importance of macroporosity on solute diffusion (Schaefer et al., 1997; Moldrup et al., 2000, 2001). Using 144 undisturbed soils of different textures, Moldrup et al. (2000, 2001) consistently observed an exponential rise in the gaseous Fickian diffusion coefficient with increase in the macropore fraction. A similar trend is expected for aqueous solute diffusion (Olesen et al., 2001a,b). By increasing access to the matrix interior, macropores can reduce diffusion path lengths, and thus the resistance (Schaefer et al., 1997) or impedance (Olesen et al., 2001a) to intra-ped solute diffusion.

The influence of macropores on the rapid transport of water and chemicals has also been extensively observed in subsurface-drained agricultural field soils (Fortin et al., 2002; Kladvko et al., 2001; Wallach and Steenhuis, 1998; Kladvko et al., 1991; Kladvko et al., 1999), with solute arrival times often being orders of magnitude less than predicted by traditional advection–dispersion models (Richard and Steenhuis, 1988; Kung et al., 2000b; Vogel et al., 2000; Kladvko et al., 2001).

When preferential transport is coupled with solute diffusion between micropore and macropore regions, a state of physical and/or chemical nonequilibrium ensues, reducing the apparent retardation (sorption) and attenuation (degradation) capacity of the biogeochemical processes in the flow domain and distorting the solute breakthrough profile. The preferential transport and subsequent nonequilibrium manifestation is controlled by the activation of all, or parts, of the macropore network, which depends on pore size, continuity, spacing, and the level of water saturation of the soil (Beven and Germann, 1982). When the macropore network is fully active (i.e. saturated),

nonreactive and reactive (i.e. sorbing) solutes can show simultaneous breakthrough characteristics, whereas the influence of chemical properties on solute transport becomes more apparent as the water saturation level of the soil decreases (Kung et al., 2000a).

The purpose of this study was to evaluate the role of macropore networks on intra-ped diffusive mass transfer and on preferential advective transport. Macropore influences on solute movement were first assessed for soil aggregates in a batch experiment and then in solute transport experiments conducted in a large, intact soil column. The objectives of the batch and column studies were to: a.) show how macropore networks influence solute diffusion, and b.) show how macropore networks affect advective transport of nonreactive and reactive solutes at different levels of water saturation.

2. Experimental methods

2.1. Soil

The soil used in the batch and soil column experiments was taken from the upper 30 cm of the A horizon in a subsurface-drained agricultural field plot at the Water Quality Field Station, located within Purdue University's Agronomy Center for Research and Education, West Lafayette, IN. The field plots are cropped with either a corn monoculture or a corn/soybean rotation. The field soil is a Drummer silty clay loam (fine-silty, mixed mesic Typic Endoquoll), has a subangular blocky structure in the upper 2 m profile, and has an A horizon extending to at least 40 cm (USDA, 1998). The soil pH is about 7.2 and the cation exchange capacity is 26.5 cmol/kg (Li and Lee, 1999). The organic carbon content of the soil is about 2% (based on measurements conducted by L.S. Lee and R. Turco in 1989 and 2001 at the Purdue University Water Quality Field Station). Although the soil is classified as poorly drained, a well-developed network of macropores, formed by root channels, earthworm burrows, and other bio-processes, can rapidly convey water and solutes to the subsurface drain during intense rainfalls.

2.2. Batch diffusion experiment

The batch diffusion experiment was conducted using 10 soil aggregates. The aggregate samples were shaped to approximate spheres by carefully breaking the soil along natural cleavage planes to avoid smearing the soil surface. To prevent structural losses during the diffusion experiment, each of the 10 soil aggregates were wrapped in a fine, nylon pouch that was tied with cotton threading. The wrapped aggregates were saturated under vacuum with a solution of 0.01 N CaCl_2 +100 mg/l bromide (Br^-)+270 mg/l pentafluorobenzoic acid (PFBA) for approximately 24 h. The CaCl_2 solution was used to suppress colloidal dispersion and the other two solutes were used to monitor diffusion rates.

Following saturation, the aggregates were suspended in 177 ml, 0.01 N CaCl_2 solution. The solution was continuously stirred with a magnetic stir-bar to create a completely

mixed system, and the solution concentration was monitored by taking 0.5 ml aliquots at frequent intervals over an 18-h period.

After the diffusion experiment was completed, the wet and dry weights and volumes of the soil (minus wrapping) were recorded to compute the average aggregate radius (1.15 cm), porosity (0.48), and bulk density (1.35 g/cm³). The average radius, r_{eff} , was computed using:

$$r_{\text{eff}} = \left(\frac{3V_T}{4N\pi} \right)^{\frac{1}{3}} \quad (1)$$

where V_T is the bulk volume of the N aggregates (Rao et al., 1982).

A few adjustments were made in processing the raw data. During the diffusion experiment, it was noticed that a small amount of water condensed along the sides and the covered top of the glass beaker. To account for this water loss, the final weight of the beaker was recorded, from which the time-averaged evaporation rate was computed. Then, the new dilution volume resulting from water losses due to sampling and evaporation was updated for each sampling interval. In addition, the measured initial concentration in the soil aggregate, C_{ao} , of the PFBA solution resulted in equilibrium concentration, C^∞ , values that overestimated the observed data. This discrepancy was likely a problem with the analysis of PFBA at the higher C_{ao} concentration, and was accounted by adjusting the C_{ao} values such that the computed C^∞ value matched the observed C^∞ value.

To further investigate the macropore influences on solute diffusion, several other soil aggregates were taken from the same location as the aggregates used in the diffusion study. The initially air-dry aggregates were wrapped in a cotton mesh to preserve structure and placed in a methylene-blue solution inside a vacuum chamber. Vacuum was applied to the chamber just until bubbling ceased from the soil aggregates. This permitted macropore voids to become saturated with the dye solution without allowing the dye to move far into the soil matrix. The soil was then removed from the dye solution, unwrapped, and oven dried for several hours to enhance the dye stains. After drying, the aggregates were broken along cleavage planes and photographed.

2.3. Transport experiments

The transport experiments were conducted in an intact soil core that was collected by inserting a stainless-steel cylinder (diameter=15 cm, length=30 cm) into the soil until the top of the cylinder was flush with the soil surface and then excavating the soil around the cylinder. The bottom of the column was carefully evened to avoid smearing, following which the bottom was capped with a stainless steel end plate containing a 0.32-cm-thick, stainless steel porous plate (air entry=70 cm H₂O, Mott Corp, USA) sandwiched between a viton 'O'-ring and viton gasket. A thin layer of sand was also placed between the soil and the porous plate to facilitate contact between the two media. Finally, a stainless steel endcap was placed on the top of the soil column and was clamped with the bottom endplate to seal the bottom of the soil column.

Table 1

Experimental boundary conditions for the seven steady flow column experiments

Water flux (cm/h)	Constant boundary conditions		Solute pulse		Solutes applied	Mean pressure head ^a (cm)
	Top	Bottom	Hr	PV		
5.66 ^b	flux	head (–3.1 cm)	0.92	0.45	Br [–] , alachlor	0.60
5.04 ^b	head	head (–3.1 cm)	0.63	0.27	Br [–] , alachlor	–0.1
2.00	flux	head (–3.1 cm)	3.00	0.53	Br [–]	–0.6
1.14	flux	head (–12 cm)	3.67	0.37	alachlor	–5.4
0.89	flux	head (–12 cm)	5.12	0.39	alachlor	–6.8
0.86	flux	head (–12 cm)	4.78	0.36	Br [–]	–7.0
0.49	flux	head (–22 cm)	7.00	0.31	Br [–] , alachlor	–15

^a Mean pressure head value estimated from HYDRUS-2D simulations.^b Flow interruptions in these experiments not considered in HYDRUS-2D simulations.

Eight transport experiments were conducted in the soil column. The first seven experiments were run as pulse applications of bromide (Br[–]) and/or alachlor herbicide (alachlor) at different flow rates and saturation levels. Constant head or flux boundary conditions were maintained in all seven experiments (see Table 1 for specific flow rates, pulse widths, and boundary conditions). The top boundary condition was established with either a small peristaltic pump (for constant flux experiments, Cole-Parmer Masterflex console drive, Model No. 7519-05) or a Mariotte reservoir (for constant head experiments). A constant negative head was maintained on the bottom boundary by attaching stainless steel tubing of the desired length to the stainless steel endplate. Steady-state flow conditions were established prior to each experiment, and outflow rates, q , were monitored prior to, and throughout, the experiments by means of volumetric sampling with an automatic fraction collector (ISCO Retriever II) to ensure that steady flow was maintained. For the $q=5.66$ cm/h and the $q=5.04$ cm/h experiments (Table 1), the flow was briefly interrupted near the end of the experiment to observe mass transfer effects. However, in the numerical simulations, the experiments were modeled without the flow interrupts, and the flow interrupt times, as well as the short rise in effluent concentrations after the interrupt period, were not considered. In the $q=0.49$ cm/h experiment, the solute residence time was sufficiently long that some degradation of alachlor was probable; however, no analyses was conducted for alachlor degradation products.

For the eighth experiment (not shown in Table 1), the bottom porous endplate was replaced with a stainless-steel perforated disk to establish a free drainage boundary condition. A nylon fabric was placed over the inner side of the disk to prevent loss of soil particles from the column. One week prior to the beginning of the experiment, the column was slowly saturated from the bottom with a 20 mg/l Br[–] solution. After saturation, a steady flow of the 20 mg/l Br[–] solution was established with a constant head of 0.4 cm at the top of the column, and the flow rate and Br[–] concentration in the effluent was monitored. A day before the start of the experiment, the column was plugged at the bottom. Several hours before the experiment, the plug was removed, steady flow reestablished, and the effluent concentration was measured. Upon observing no change in the effluent concentration, it was assumed that the entire column (macropores and soil matrix) was fully saturated with the 20 mg/l Br[–] solution. At the beginning of the

experiment ($t=0$), the influent Br^- reservoir was quickly switched to a solution containing 20 mg/l of PFBA to allow the PFBA solution to displace the resident Br^- solution. To evaluate the mass transfer dynamics, the flow was interrupted twice by plugging the column at the bottom during the course of the experiment. The first flow interrupt started at $t=3$ h and lasted for 1.9 h, while the second flow interrupt started at $t=6.9$ h and lasted for 8.9 h.

2.4. Chemical analyses

Bromide, PFBA, and alachlor were analyzed using a Shimadzu HPLC with a UV detector. Injection volumes were 10 μl for all analyses. Bromide was measured using a Dionex AS14 ion exchange column (wavelength, λ , =205 nm) and a 10 mM sodium borate mobile phase. PFBA was measured using a deactivated reverse phase C18 column (ABZ+, SUPELCO, Sigma-Aldrich), λ =205 nm, and a 90/10 v/v acetonitrile/water mobile phase with pH adjusted to 3.5 using dilute HCl. Retention times were 3.22 min for Br^- and 2.46 min for PFBA. Alachlor was analyzed with a deactivated reverse phase C18 column (ABZ+, SUPELCO, Sigma-Aldrich), λ =245 nm, and a 50/50 v/v acetonitrile/water mobile phase. The retention time for alachlor was 8.65 min.

3. Analytical and numerical modeling

3.1. Simulating intra-ped diffusion in batch experiments

Intra-ped solute diffusion from the soil aggregates in the batch experiments was modeled with Fick's Second Law for radial diffusion:

$$\frac{\partial C}{\partial t} = D_a \left(\frac{\partial^2 C}{\partial r^2} + \frac{2}{r} \frac{\partial C}{\partial r} \right) \quad (2)$$

where C is the solution concentration, t is time, r is the diffusion path radius, and D_a is the apparent Fickian diffusion coefficient. The analytical solution for the average concentration in the spheres, $\bar{C}_A$, for the situation of diffusion from initially saturated spheres to a closed, aqueous system of finite volume that is initially free of solutes is given by:

$$\bar{C}_A(t) = C^\infty + (C_A^0 - C^\infty) \sum_{n=1}^{\infty} \left\{ \left[\frac{6\beta(\beta+1)}{9+9\beta+q_n^2\beta^2} \right] \exp \left(-\frac{D_a q_n^2 t}{a^2} \right) \right\} \quad (3)$$

and the concentration in the bulk solution, C_M , is:

$$C_M(t) = C_M^0 + \frac{\theta_A}{\theta_M} [C_A^0 - \bar{C}_A(t)] \quad (4)$$

where C^∞ is the equilibrium concentration and other terms are as given in Appendix A (also see the Appendix B for further details of the above solution).

3.2. Simulating preferential transport in a column system

All numerical simulations of the column experiments were conducted using HYDRUS-2D (Simunek et al., 1999a) modified to include a dual-porosity module (Simunek et al., 2003). The dual-porosity (i.e. mobile–immobile) model accounts for both advective water movement and solute diffusion between the mobile (macropore) and immobile (matrix) regions. Intra-ped transfer for advection and diffusion are modeled as first-order processes driven by the pressure and concentration gradients, respectively. The equations for water and solute for the dual-porosity model are (Simunek et al., 2003):

Water:

$$\frac{\partial \theta_f}{\partial t} = \frac{\partial}{\partial z} \left(K_f \frac{\partial h_f}{\partial z} - K_f \right) - \Gamma_w \quad (5a)$$

$$\frac{\partial \theta_m}{\partial t} = \Gamma_w \quad (5b)$$

$$\Gamma_w = \alpha_w (h_f - h_m) \quad (5c)$$

Solute:

$$\frac{\partial}{\partial t} \left(\theta_f R_f C_f \right) = \frac{\partial}{\partial z} \left(\theta_f D_f \frac{\partial C_f}{\partial z} - q_f C_f \right) - \Gamma_s \quad (6a)$$

$$\frac{\partial}{\partial t} \left(\theta_m R_m C_m \right) = \frac{\partial}{\partial z} \left(\theta_m D_m \frac{\partial C_m}{\partial z} \right) - \Gamma_s \quad (6b)$$

$$\Gamma_s = \Gamma_w C^* + \alpha_D (C_f - C_m) \quad (6c)$$

where the subscript f denotes the macropore (fracture) domain, the subscript m the matrix domain. Also, C is the aqueous solute concentration, θ is the volumetric water content, h is the matric potential, R is the retardation factor, D is the dispersion coefficient, $C^*=C_f$ for $\Gamma_w > 0$ and C_m for $\Gamma_w < 0$, and α_w and α_D are the first-order rate constants for inter-domain transfer of water and solute, respectively (other nomenclature is as given in Appendix A). Because all column experiments were conducted under saturated and/or unsaturated steady-state water flow conditions, there was no pressure-head gradient between the matrix and macropore regions. Hence, diffusive transfer was assumed to be responsible for all intra-ped solute exchange.

Eqs. (5a)–(5c), (6a)–(6c) only account for nonequilibrium due to mass transfer between the mobile and immobile domains. This representation is sufficient to describe the nonequilibrium transport of a nonreactive solute such as Br^- , but may not be adequate to describe the nonequilibrium transport of alachlor, which can be affected by both mass

transport constraints (physical nonequilibrium) and slow sorption–desorption kinetics (sorption nonequilibrium). The HYDRUS-2D code also includes a sorption nonequilibrium model that uses equations analogous to Eqs. (5a)–(5c), (6a)–(6c) with instantaneous and kinetic sorption domains instead of mobile and immobile domains and a first-order desorption rate constant instead of a first-order diffusion term (Brusseau and Rao, 1989; Simunek et al., 1999a). However, both nonequilibrium models cannot be used simultaneously in HYDRUS-2D. Some of the simulations of alachlor transport were made using the physical nonequilibrium model and others were made with the sorption nonequilibrium model, depending on which model provided a better fit to the experimental data. Hence, the nonequilibrium parameters calibrated for the alachlor transport studies are difficult to physically interpret because they lump the effects of all nonequilibrium processes affecting alachlor transport.

The simulations for the seven steady flow experiments with Br^- and alachlor were terminated after a 97% mass recovery. The slight truncation of the simulated BTCs was necessary because of the lengthy run-times needed in order to generate the full concentration profiles.

4. Parameter estimation

4.1. Effective diffusion radius

The two unknown parameters in Eq. (3) are the Fickian diffusion coefficient, D_a , and the effective diffusion radius of the spheres, a . The values of D_a for Br^- and PFBA were independently estimated using a bulk water diffusion coefficient, D_o , of $0.0648 \text{ cm}^2/\text{h}$ for Br^- (Moldrup et al., 2003) and $0.024 \text{ cm}^2/\text{h}$ for PFBA (estimated using the method of Hayduk and Laudie, 1974 as recommended in Lyman et al., 1982). To validate these bulk water diffusion coefficients, a separate batch diffusion study (data not shown) was conducted with the same ceramic spheres ($r=0.75 \text{ cm}$) used by Rao et al. (1980a) in their batch diffusion study. Because, Rao et al. (1980a) used chloride rather than Br^- or PFBA in their experiment, the observed D_a of the Rao et al. (1980a) study was related to the D_a values observed here for Br^- and PFBA using the impedance factor relationship (D_a/D_o), which is only a function of the porous system network (e.g. tortuosity, water content) and not the solutes (Olesen et al., 2001a). The impedance factor measured by Rao et al. (1980a,b) was 0.30, and the impedance factors measured here were 0.32 for Br^- and 0.29 for PFBA. The good agreement between experiments provided confidence in the repeatability of the experimental protocol as well as reliable estimates of the D_o values for Br^- and PFBA.

After validating the D_o values for the solutes, the D_a value for the solutes in the field soils could also be estimated from the impedance factor relationship. The Buckingham (1904) model (as suggested by Moldrup et al., 2003 for soils at water saturation) was used to estimate the ideal impedance factors for Br^- and PFBA in the field soils. The Buckingham model is:

$$\frac{D_a}{D_o} = (\theta_T)^2 \quad (7)$$

where θ_T is the total water content of the soil (equal to the soil porosity when the soil is saturated). The measured soil porosity, and estimated D_o values, the Buckingham model predicted $D_a=0.015$ cm²/h for Br[−] and 0.006 cm²/h for PFBA. With the independent estimation of the D_a values, the effective diffusion radius for the soil aggregates could be found via minimizing the squared residuals between the measured and computed C_M values using Eqs. (3) and (4). Using this method, Rao et al. (1980a) showed the estimated diffusion radius to be equal to the measured radius of their homogeneous ceramic spheres. This result was confirmed in the repeat of their experiment conducted in this study.

4.2. Soil hydraulic parameters

To avoid concerns of parameter nonuniqueness, soil hydraulic properties for the column were either independently measured or estimated from additional column experiments. The saturated hydraulic conductivity (K_s) was computed from the constant-head experiments and ranged from 2.0 to 5.7 cm/h over the course of the experiments. A total soil porosity (θ_T) of 0.41 was estimated from porosity measurements of several soil cores taken at field sites near the water quality field station (unpublished data) and is close to the porosity measured for the soil aggregates used in the batch diffusion experiments. The van Genuchten (1980) water retention parameters (α , n) were estimated using the HYDRUS-2D inverse module (Simunek et al., 1999b) for a separate water drainage experiment (mobile: $\alpha=0.04$, $n=1.4$; immobile: $\alpha=0.005$, $n=1.5$). In the numerical simulations for Br[−], the effective mobile domain saturated water content, θ_{sf} , and the first-order diffusion term, α_D , were the only calibrated parameters (θ_{sm} was computed as $\theta_{sm}=\theta_T-\theta_{sf}$), and these parameter values were used to gain insight into the macropore fraction and diffusion lengths in the soil column. In addition to using HYDRUS-2D to find θ_{sf} and α_D values for Br[−], the pulse-corrected first moment of the simulated BTCs were also used to estimate the pore volume of the soil column for each experiment.

In the numerical simulations of alachlor, θ_{sf} and α_D were treated as fitting parameters if the physical nonequilibrium model was used, and θ_{sf} , K_d (sorption partition coefficient), f (fraction of instantaneous sorption), and k_2 (desorption rate constant) were the fitting parameters if the sorption nonequilibrium model was used. Because the fitted parameters for alachlor lumped different nonequilibrium processes, no physical meaning was attached to their calibrated values. The simulated results for alachlor were used to further complete the breakthrough profiles for alachlor and compute the pulse corrected moments, but not to make inferences on diffusion lengths or the macropore fraction.

4.3. First-order mass transfer parameters for bromide simulations

The values of the first-order mass transfer coefficient, α_D , calibrated in the Br[−] simulations were converted to an equivalent Fickian diffusion coefficient using the solution outlined in Rao et al. (1980b), which accounts for the effects of the solute

mean retention time and cyclic mass transfer on the diffusion exchange term (Young and Ball, 1995, 1997; Griffioen, 1998):

$$\alpha_D = \frac{D_a \varphi \theta_A}{a^2} \frac{\sigma_1}{\sigma_2} \quad (8)$$

where σ_1 and σ_2 are time-dependent correction factors to account for incomplete diffusion effects and φ is the mobile fraction (see Appendix A).

Rao et al. (1980a) further reasoned that the solute mean travel time is a good approximation of mean solute contact time and showed that the time-averaged α_D can be expressed as:

$$\bar{\alpha}_D = \frac{D_a \theta_A \varphi q_1^2}{a^2 (1-b)} \left[\frac{0.1}{T^*} \right]^b, \quad 10^{-4} < T^* \leq 0.1 \quad (9a)$$

$$\bar{\alpha}_D^* = \frac{D_a \theta_A \varphi q_1^2}{a^2} \left[1 + \frac{0.1b}{(1-b)T^*} \right], \quad T^* > 0.1 \quad (9b)$$

where

$$b = 0.11472 \ln \left(\frac{167}{\varphi q_1^2} \right) \quad (9c)$$

and T^* is the nondimensionalized time, $T^* = D_a t / a^2$.

In addition to the experimental time, the geometry of the soil's matrix-macropore structure also affects the translation of the first-order diffusion term to a Fickian diffusion coefficient. A variety of methods have been used to analytically derive shape factors to convert the diffusion path length based on a spherical geometry to an equivalent path length for other geometric shapes (see Parker and Valocchi, 1986; van Genuchten and Dalton, 1986; Valocchi, 1990; Gerke and van Genuchten, 1996). van Genuchten and Dalton (1986) demonstrated that values of α_D derived for a spherical geometry (S) can be converted to equivalent values of α_D for non-spherical geometries (NS) using nondimensional shape factors (κ):

$$\frac{\kappa_{NS}}{a_{NS}^2} = \frac{\kappa_S}{a_S^2} \quad (10a)$$

or

$$a_{NS} = \left(\frac{\kappa_{NS}}{\kappa_S} \right)^{\frac{1}{2}} a_S \quad (10b)$$

Several methods have been used to analytically derive shape factors, including matching second temporal moments (Parker and Valocchi, 1986; Valocchi, 1990), comparing Laplace transforms (van Genuchten and Dalton, 1986), and directly matching average concentration curves (van Genuchten and Dalton, 1986; Gerke and van Genuchten, 1996). Most of these methods estimate shape factor values of 3 for rectangular slabs, 8 for solid cylinders, and 15 for spheres. The subangular blocky structure of the soil

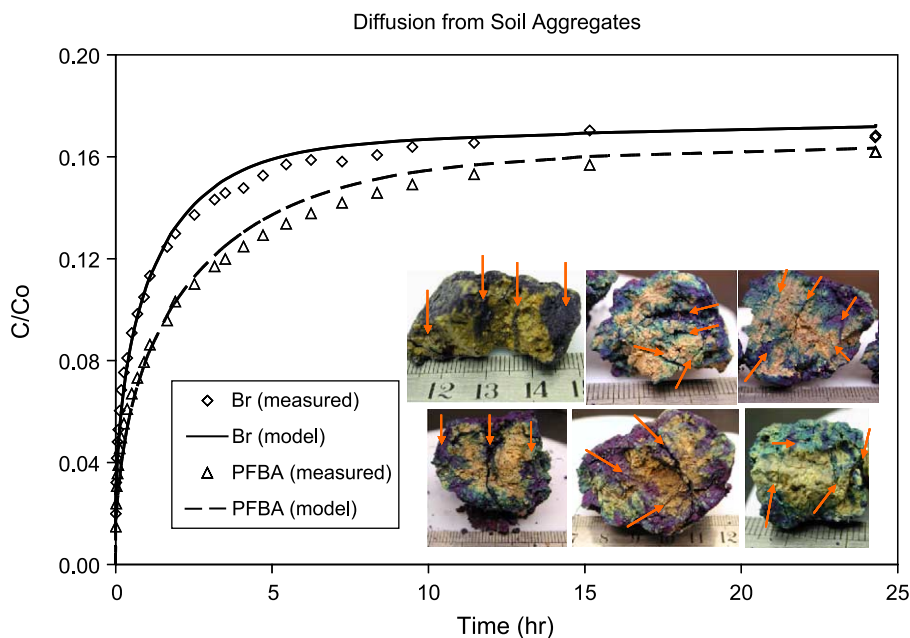


Fig. 1. Measured and modeled aqueous concentrations of bromide (Br^-) and pentafluorobenzoic acid (PFBA) in the batch diffusion experiment. The photo shows dye-stained soil aggregates revealing cracks and macropores (mm scale).

used in these experiments was approximated as a rectangular slab, giving a conversion ratio of 0.45 for translating from a spherical geometry to a blocky geometry.

5. Results and discussion

5.1. Batch diffusion experiments

The aqueous concentrations of Br^- and PFBA measured in the batch diffusion experiments rapidly rise and reach their equilibrium concentrations in about 10 h (Fig. 1). The slower rise in the PFBA concentration compared to that of Br^- is attributed to the larger bulk diffusion coefficient of PFBA. The observed data for both solutes are modeled well with Eq. (4) ($R^2 > 0.99$). The fitted value for the effective diffusion radius (a) of the soil aggregates is 0.54 cm for the Br^- data and 0.46 cm for the PFBA data. These estimated values of a are less than half the measured mean radius (r_{eff}) of the soil aggregates (1.15 cm).

The observation of $a < r_{\text{eff}}$ connotes that the soil macropores provide increased access (reduced impedance) for solute diffusion into the interior of the soil matrix. This influence of macropores on a is supported by the stained soil aggregates (photo in Fig. 1). Dye stains are concentrated along ped faces and are principally oriented perpendicular to each other, indicative of the subangular blocky structure of the soil.

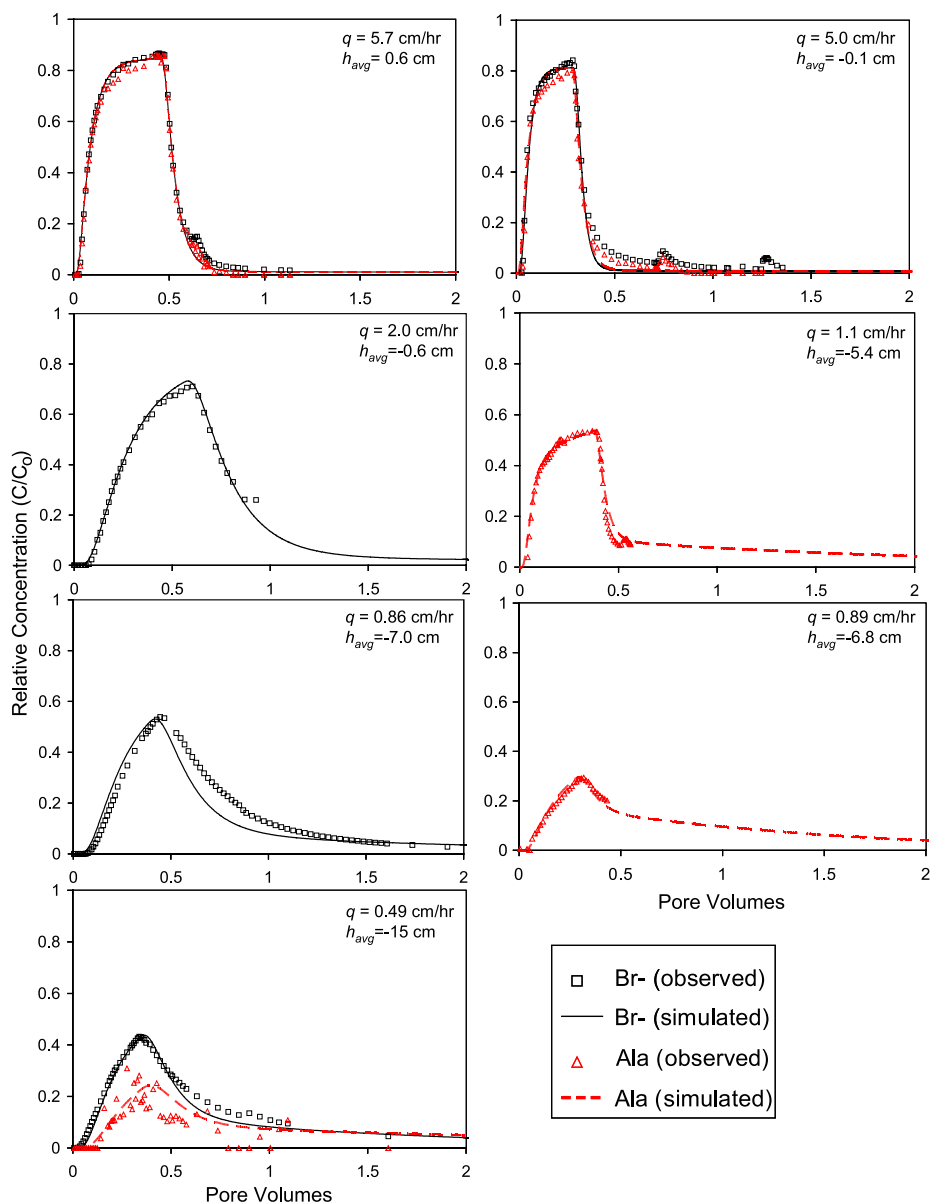


Fig. 2. Measured and simulated breakthrough curves for bromide (Br^-) and alachlor (Ala) in the seven steady flow column experiments. The concentration spikes in the tail of the measured data for the $q=5.7$ cm/h and the $q=5.0$ cm/h are flow interruption periods (not considered in the HYDRUS-2D simulations).

The mean distance between inter-ped crack faces, as highlighted by the dye stains, is roughly about 1 cm (or a crack half-spacing of about 0.50 cm). The correspondence of the fitted a values from the batch diffusion experiments to the half-spacing of the

structural interfaces between soil peds implies that the smaller intra-ped cracks, and not only the larger bio-pores, play an important role in reducing the impedance of diffusion into the soil matrix.

5.2. Transport experiments

The BTCs for both Br^- and alachlor exhibit nonequilibrium effects for all experiments conducted under steady-flow conditions (Fig. 2). In all experiments, peak arrival times occur much earlier than one pore volume, indicating significant bypass flow through preferential channels. Nonequilibrium is most evident at the higher flow rates where the BTCs for Br^- and alachlor are nearly identical. As the flow rate decreases, solute peak times are increased and breakthrough is more dispersed (Fig. 2). The more symmetric BTCs at the lower flow rates show the progressive approach to equilibrium.

In the flow interruption experiment, the initial behavior of the PFBA breakthrough is similar to the Br^- BTC of the saturated, steady-flow experiments (Fig. 3). The sharp drop in concentration following the flow interruptions confirms the bypass channeling suspected in the steady flow experiments. Also, the BTC for Br^- mirrors the PFBA breakthrough. The reciprocal BTCs for the two solutes denote that diffusion in and out of the soil matrix proceeds at the same rate; as expected, the intra-ped diffusion process is non-hysteretic.

The BTCs simulated with HYDRUS-2D match well with the measured data for all experiments (Figs. 2 and 3). Deviations are most pronounced in the tailing of the

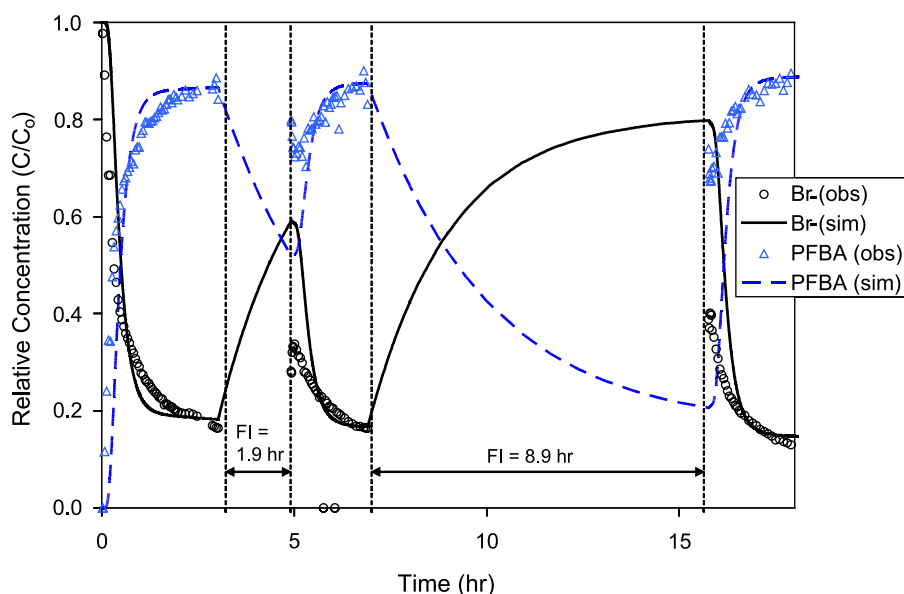


Fig. 3. Measured and simulated breakthrough curves for the soil column experiment with pentafluorobenzoic acid (PFBA) displacing bromide (Br^-) with two flow interruptions.

experiment with the flow rate of $q=5.04$ cm/h and in the initial recession for the experiment at $q=0.86$ cm/h. The discrepancies between simulated and observed mass transfer during flow interrupts of the flow interruption experiment (Fig. 3) likely arise from a 30 ml dead-space at the bottom of the column that diluted the initial solute concentrations, thus decreasing Br^- concentrations and increasing PFBA concentrations for the first several samples after each flow interruption. The simulation of the

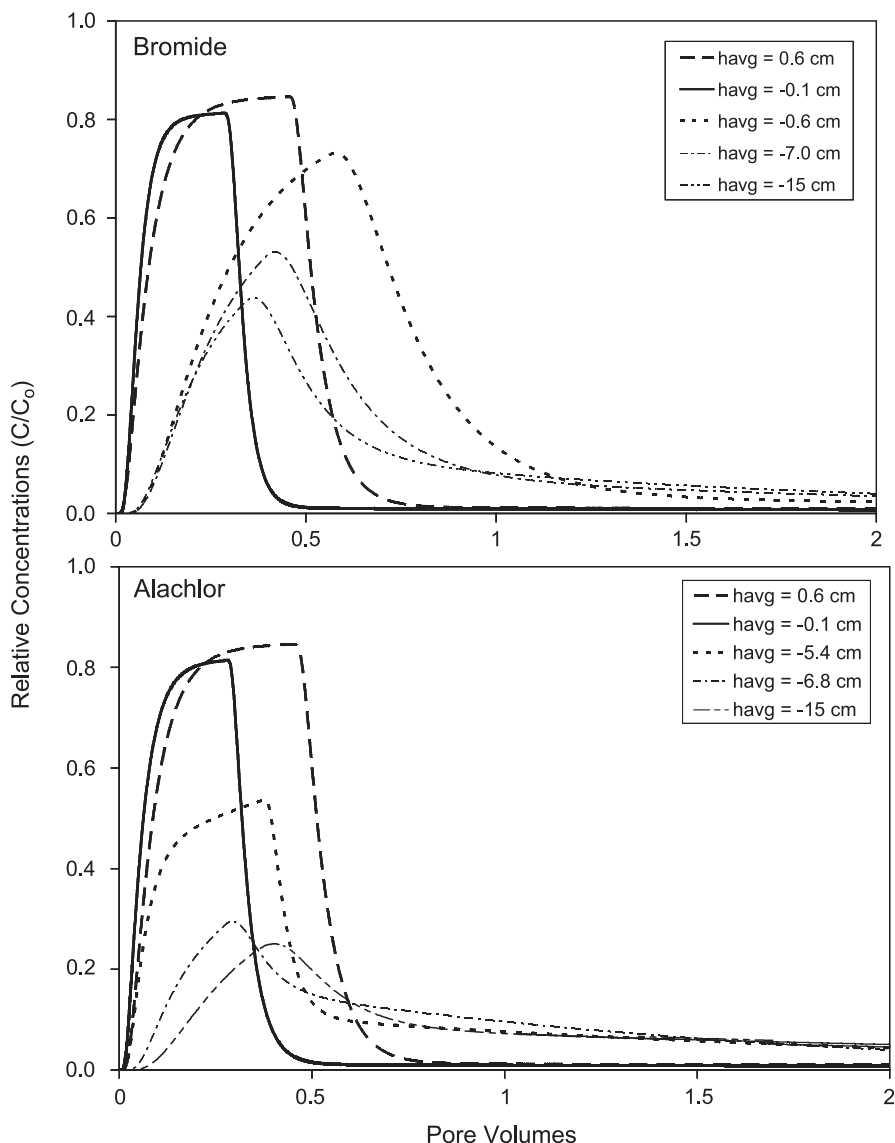


Fig. 4. Simulated solute breakthrough curves plotted against pore volumes for bromide and alachlor (see Fig. 2).

alachlor BTC for the $q=0.49$ cm/h experiment was also problematic because of poor experimental data quality. As a result, the interpretation and simulation of this particular experiment is somewhat subjective. Despite these considerations, the HYDRUS-2D simulations capture the key attributes of the BTCs. Because of the premature truncation of a few of the measured BTCs, the discussion of the soil column experiments will primarily focus on the simulated BTCs. The use of the simulated BTCs is particularly helpful in estimating the tailing of the BTCs when computing temporal moments. In addition, the simulated data provided an estimation of the mean pressure head (h_{avg}) so that each experiment could be related to the mean saturation level in the soil column. The simulated BTCs are grouped by solute type in Fig. 4.

To account for the different pulse widths used in the steady-flow experiments, the Br^- and alachlor BTCs are further analyzed using the pulse-corrected temporal moments of the simulated BTCs. Since truncated breakthrough data leads to greater errors in the temporal moments with increasing nonequilibrium (Das et al., 2002), the deviation of the computed moments derived from the HYDRUS-2D simulations truncated at 97% mass recovery relative to the ideal (equilibrium) moments (also truncated at 97% mass recovery) provides an assessment of the degree of non-equilibrium. The pulse-corrected, second, centralized moment truncated at 97% mass recovery is plotted in nondimensional time (pore volumes) in Fig. 5. Also shown are the truncated equilibrium moments for Br^- and alachlor. The alachlor equilibrium BTC was generated in HYDRUS-2D assuming a sorption coefficient, K_{oc} , of 170 ml/g (Ware, 1992).

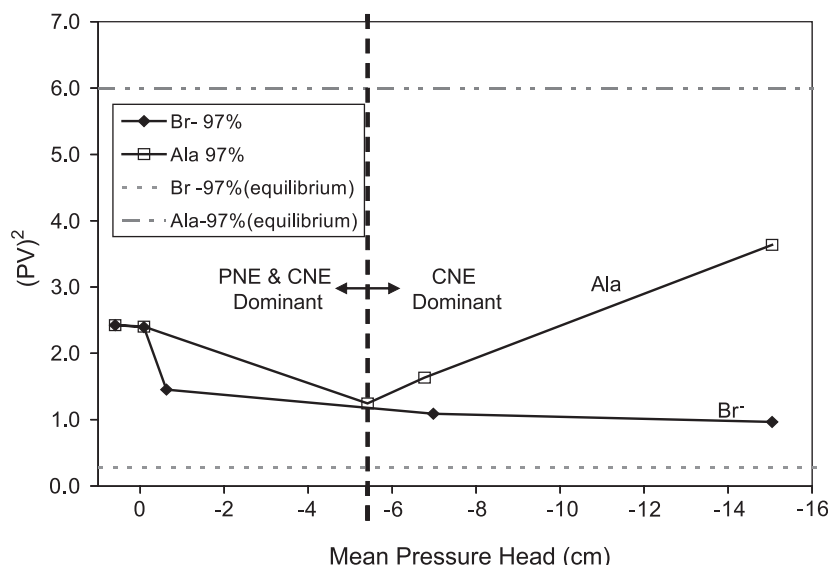


Fig. 5. Pulse-corrected, second, centralized, temporal moments for simulated (97% mass recovery) bromide (Br^-) and alachlor (Ala) BTCs. The equilibrium moments for Br^- and Ala are also plotted as horizontal lines.

Table 2

Calibrated macroporosity and diffusion mass transfer terms for flow interrupt (FI) experiments and steady-state bromide experiments

Exp. no.	Water flux (cm/h)	Mean pressure head (cm)	Mean travel time (h)	Calibrated θ_{sf}	Calibrated α_D (h ⁻¹)	Estimated diffusion radius, a (cm)
1	FI (Br ⁻)	0.5	—	0.03	0.013	0.91
2	FI (PFBA)	0.5	—	0.03	0.009	0.96
3	5.7	0.6	2.04	0.04	0.037	0.96
4	5.0	-0.1	2.29	0.028	0.040	0.84
5	2.0	-0.6	5.78	0.13	0.018	1.17
6	0.86	-7.0	13.38	0.14	0.017	0.84
7	0.49	-15	23.45	0.13	0.016	0.68

5.2.1. Diffusive mass exchange at the column-scale

The BTCs of all the column experiments show that intra-ped diffusion is a key process leading to physical nonequilibrium in dual-porosity media. As predicted from Eq. (8), the calibrated values of $\bar{\alpha}_D$ for all the Br⁻ experiments tend to decrease with longer experimental times (Table 2). The fitted values of both θ_f and α_D are well within ranges commonly reported in the literature for measurements conducted on similar field soils (Jaynes et al., 1995; Casey et al., 1997, 1998, 1999).

The values for the effective diffusion radius, a , computed from the calibrated $\bar{\alpha}_D$ values (see Eqs. (8), (9a)–(10b)) ranged from 0.68 to 1.2 cm with an average of 0.91 cm (Table 2)—nearly double the $a \sim 0.50$ cm estimated in the batch diffusion experiments. A possible explanation for larger a values in the transport simulations compared to the batch study is that structural voids, root channels, and other preferential pathways are more prone to penetrate the length of soil aggregates than in the larger soil column, and the loss of pathway continuity decreases access to the matrix interior—thus increasing the diffusion path length. Another potential cause for the increased a values is that longer transport distances at the column-scale amplify diffusion impedances in the preferential pathways. The calculations of a in the batch experiments implicitly assumed that macropores and inter-ped fissures provided no impedance to solute diffusion. This assumption was justified at the aggregate-scale because overall transport lengths were relatively short, with any macropore impedance being much less than that of the soil matrix. At the column-scale, however, transport distances through the cracks and macropores are much longer and likely more tortuous. The increased distances would magnify the effect of diffusion impedance so that the impedance through the preferential pathways, while still less than that of the soil matrix, becomes significant in the soil column system.

5.2.2. Preferential transport of nonreactive solute

The progressive shift to the right and decreased skewing of the BTCs with greater unsaturation (Fig. 4) is particularly characteristic of dual-domain media such as soil with structural macropores (Brusseau and Rao, 1990; Langner et al., 1999). Despite the progressive shift to the right, all BTCs have concentration peaks at pore volumes less

than one, an indication that some degree of physical nonequilibrium is manifest at all saturation levels.

The BTCs for Br^- show two trends in the solute breakthrough characteristics (Fig. 4). For the two experiments conducted at the highest saturation levels, solute breakthrough occurs almost immediately, rapidly rising to over 80% of the input concentration (the apparent earlier arrival of solute in the $h_{\text{avg}}=-0.1$ cm experiment compared to the $h_{\text{avg}}=0.6$ cm experiment likely arises from differences in the way the initial time was determined for a constant head boundary condition ($h_{\text{avg}}=-0.1$ cm experiment) versus a constant flux boundary condition ($h_{\text{avg}}=0.6$ cm experiment)). These two BTCs are highly skewed, with an almost step change in the concentration profile. Such behavior is characteristic of an advection-dominated transport regime that provides little time for interaction with the soil matrix. The BTCs for the three other experiments conducted at higher matric potentials still show significant preferential transport, but they have initial and peak breakthrough times that are slightly later, and concentration profiles that are more symmetric, than the two higher saturation experiments.

The two trends in the BTCs indicate that two types of preferential transport networks operate in the soil column. For pressure levels at or near the saturation pressure, the solutes are rapidly flushed through a network of large macropores that dominate all other flow and transport processes. At slight tensions (i.e. greater than $h_{\text{avg}}=-0.6$ cm) these large macropores drain, allowing a secondary network of smaller voids (yet large enough to permit preferential channeling) to dominate the transport of solutes.

The values of the θ_{sf} calibrated for Br^- in the HYDRUS-2D simulations (Table 2) provide complementary indication of primary and secondary preferential networks. For the experiments at or near saturation ($h_{\text{avg}} \geq -0.1$ cm), θ_{sf} is very small ($0.028-0.04 \text{ cm}^3/\text{cm}^3$), whereas the simulations at greater unsaturation levels ($h_{\text{avg}} < -0.1$ cm) have a larger θ_{sf} value ($0.13-0.14 \text{ cm}^3/\text{cm}^3$). These θ_{sf} values seem counterintuitive (i.e. higher matric potentials should correspond to decreasing water contents). Apparently, at the two highest flow-rates/saturation levels, even though water fills a larger number of pores, the dominant flow dynamics occur in the larger macropores that occupy only 3–4% of the transport domain. When the flow rate (saturation level) decreases slightly, the larger macropores become inoperative and transport occurs predominantly in a network of smaller macropores whose air-entry pressure has not yet been exceeded at tensions between 0.6 and 15 cm. These smaller macropores occupy a water content of 0.13 to 0.14 cm^3/cm^3 , a fraction greater than the primary macropore network but still significantly less than the total porosity (0.41).

The second moment for the truncated Br^- BTCs plotted against the mean column pressure head (Fig. 5) rapidly drops to approach its equilibrium value at the $h_{\text{avg}}=-0.6$ cm pressure head. Despite the steep decrease in the value of the second moment at $h_{\text{avg}}=-0.6$ cm, the truncated second moments suggest that some degree of physical nonequilibrium is still present even at a potential of 15 cm. The drop in the truncated second moment in the $h_{\text{avg}}=-0.1$ cm experiment and the $h_{\text{avg}}=-0.6$ cm experiment (Fig. 5) further points to a primary network that dominates the flow regime near saturation and a secondary preferential network that continues to operate after the primary network drains.

The equivalent size of the macropores participating in preferential flow can be related to the corresponding mean pressure head with the capillary rise equation (Domenico and Schwartz, 1998, p. 394):

$$r = \frac{2\sigma\cos\gamma}{\rho_w g \psi} \quad (11)$$

where σ is the surface tension for water (0.072 N/m at 25 °C), γ is the water–solid contact angle (assume to be 0°), ρ_w is the density of water (1000 kg/m³ at 25 °C), g is the gravitational constant (9.81 m/s²), and ψ is the matric potential. For potentials of $\psi=0.6$ cm and $\psi=15$ cm, the computed macropores size is approximately 2400 and 100 μ m, respectively (though pore structure and continuity are also important criteria in the expression of macropore flow; Beven and Germann, 1982). Given the data in this study, it appears that continuous pore sequences with radii in the order of 100 μ m are sufficiently large to lead to preferential transport (i.e. physical nonequilibrium). However, the preferential transport in these pores is dominated by transport through larger pore networks ($r\sim 2400$ μ m) that occupy a much smaller fraction of the soil volume. Though not verified in this study, it is surmised from visual observations of the soil that the primary macropore network ($r\sim 2400$ μ m) is composed of bio-pores such as earthworm holes and root channels; whereas the secondary network ($r\sim 100$ μ m) is constructed of the intra-ped fissures inherent in the soil structural units.

5.2.3. Preferential transport of reactive solute

For the two steady-state experiments conducted at the highest flow rates ($q=5.66$ cm/h and $q=5.04$ cm/h), the alachlor BTCs are impressively similar to those of Br[−] (Fig. 2). The simultaneous breakthrough of reactive solutes with nonreactive tracers has been cited in several studies as a remarkable, yet characteristic, display of preferential flow (Kung et al., 2000a; Kladienko et al., 2001; Fortin et al., 2002). Though the peak breakthrough times of alachlor and Br[−] are similar for all experiments, the alachlor BTCs exhibit more asymmetry with longer tails at the higher matric potentials. These BTC characteristics are highlighted in the graph of the truncated second moments for the Br[−] and alachlor BTCs. Near saturation, such a high degree of nonequilibrium (physical and chemical) transport occurs that the Br[−] and alachlor moments are virtually identical. As physical nonequilibrium starts to dissipate under slightly greater tensions, the moments for Br[−] and alachlor begin to diverge. The reduction of physical equilibrium leads to a decrease in the second moment for both Br[−] and alachlor between the $h_{\text{avg}}=0.60$ cm and $h_{\text{avg}}=-5.4$ cm experiments. At experiments with $h_{\text{avg}}<-5.4$ cm, the second moment for Br[−] continues to decrease, while the second moment for alachlor begins to increase (due to the increasing apparent retardation of alachlor). By the $h_{\text{avg}}=-15$ cm experiment, the truncated second moment for Br[−] is near its equilibrium value while the second moment for alachlor is still far below its equilibrium value. If all nonequilibrium behavior for Br[−] can be attributed to physical nonequilibrium, then differences in the nonequilibrium behavior of alachlor relative to the nonequilibrium behavior of Br[−] result from sorption nonequilibrium in the alachlor transport. Based on this logic, it appears that while physical equilibrium is approached in the more unsaturated

experiments, the alachlor transport still exhibits a considerable degree of sorption nonequilibrium at the lower flow rates. Thus, even a slight amount of macropore channeling (i.e. physical nonequilibrium) can lead to a high degree of transport nonequilibrium (sorption+physical) for alachlor.

6. Summary and conclusions

Macropore networks in agricultural field soils influence both intra-ped solute diffusion and transport through the soil profile. The coupling of the two processes results in reduced peak breakthrough times and extended tailing of breakthrough curves. Macropore networks permeating the soil matrix increase access to the ped interior, effectively reducing the diffusion radius and increasing the solute diffusion rate. The estimated diffusion radius of the batch diffusion experiment with several soil peds roughly corresponds to the half-spacing of intra-ped cracks, suggesting that these smaller voids, and not only the larger bio-pores, are sufficiently large to reduce the impedance of solute diffusion into the ped interior.

In the experiments with the intact soil column, the diffusion radius is nearly twice that from the batch experiments. The increase in the effective diffusion path length for the column versus batch experiments is attributed to an increased tortuosity that results from less contiguous preferential networks and the increased importance of the diffusion impedance through macropores that was neglected in the batch experiment.

The preferential transport of a nonreactive solute (bromide) and a reactive solute (alachlor) was assessed using simulated BTCs fitted to the experimental column studies that were conducted under different saturation levels. The preferential flow and nonequilibrium conditions were more apparent with increasing flow rates and reveal two types of preferential networks. Near saturation, large macropores ($r \sim 2400 \mu\text{m}$) that occupy only about 3–4% of the total domain dominate the flow and transport response. When these larger pores drain under slight negative pressures, a lesser degree of physical nonequilibrium still persists due to a secondary network of smaller ($r \sim 100 \mu\text{m}$), yet more numerous, macropores. It is likely that the primary network of larger macropores is composed of the earthworm burrows and root channels, and the secondary network is the intra-ped voids inherent in the soil structure.

The alachlor breakthrough curve exhibited significant nonequilibrium effects even as the nonequilibrium in the bromide breakthrough appeared to be dissipating. The continued tailing and early breakthrough of the alachlor is attributed to sorption nonequilibrium and illustrates that even a slight amount of preferential channeling may lead to a considerable degree of chemical nonequilibrium for reactive solutes.

This study illustrates how dual-domain (macropore-soil matrix) networks influence solute movement, both diffusion and advective transport, at two observation scales. The differences in the dual-domain characteristics in translating from the batch-scale to the column-scale are likely compounded when upscaling to a field system where macropore networks operate in a multi-dimensional domain and interact with larger-scale soil heterogeneities. Describing the preferential networks and predicting solute transport at the field-scale are the subjects of further investigations.

Appendix A. Notation

a	mean diffusion radius (L)
h_{avg}	mean pressure head in soil column (L)
h_f	mean pressure head in the macropore (mobile) domain (L)
h_m	mean pressure head in the matrix (immobile) domain (L)
n	water retention parameter from van Genuchten (1980) model
q_n	roots of the equation $\tan q_n = 3q_n / (3 + \beta q_n^2)$
q_f	water flux density in the macropore (mobile) domain (L/T)
r	radial coordinate of a sphere (soil ped) (L)
r_{eff}	average radius of soil aggregates used in batch diffusion experiments (L)
t	time (T)
z	length coordinate for one-dimensional soil column (L)
C_f	concentration in the macropore (mobile) domain (M/L ³)
C_m	concentration in the matrix (immobile) domain (M/L ³)
C^*	C_f for $\Gamma_w > 0$ and C_m for $\Gamma_w = 0$ (M/L ³)
C_A	concentration within a sphere (soil ped) (M/L ³)
C_A^0	initial ($t=0$) concentration in a sphere (soil ped) (M/L ³)
C_M	concentration in the solution external to a sphere (soil ped) (M/L ³)
C_M^0	initial ($t=0$) concentration in the solution external to a sphere (soil ped) (M/L ³)
C^∞	equilibrium concentration ($t \rightarrow \infty$) for the batch system (M/L ³)
D_a	effective diffusion coefficient for solute in porous media (L ² /T)
D_f	dispersion coefficient in the macropore (mobile) domain (L ² /T)
D_m	dispersion coefficient in the matrix (mobile) domain (L ² /T)
D_o	bulk diffusion coefficient for solute in water (L ² /T)
K_f	hydraulic conductivity of the macropore (mobile) domain (L/T)
M_T	total solute mass per unit volume of system (M)
N	number of soil aggregates
R_f	Retardation factor in the macropore (mobile) domain
R_m	Retardation factor in the matrix (mobile) domain
V_A	total volume of solution within spheres (soil peds) (L ³)
V_M	total volume of solution external to the spheres (soil peds) (L ³)
V_T	total volume occupied by the soil aggregates in the batch diffusion experiment (L ³)
α	water retention parameter from van Genuchten (1980) model (1/L)
α_D	first-order solute diffusion rate constant (1/T)
$\bar{\alpha}_D$	time-averaged value of α_D (1/T)
α_w	first-order water transfer rate constant (1/(LT))
β	θ_M / θ_A
κ	geometry shape factor for the first-order solute diffusion term, α_D
σ_1	$\sum_{n=1}^{\infty} \left\{ \left[\frac{6\beta(\beta+1)q_n^2}{9+9\beta+q_n^2\beta^2} \right] \exp \left(-\frac{D_a q_n^2 t}{a^2} \right) \right\} \text{ (Crank, 1975)}$
σ_2	$\sum_{n=1}^{\infty} \left\{ \left[\frac{6\beta(\beta+1)}{9+9\beta+q_n^2\beta^2} \right] \exp \left(-\frac{D_a q_n^2 t}{a^2} \right) \right\} \text{ (Crank, 1975)}$
Γ_s	solute transfer term = $\Gamma_w C^* + \alpha_D (C_f - C_m)$ (M/L ³ T)
Γ_w	water transfer term = $\alpha_w (h_f - h_m)$ (1/T)
θ_f	water content in the macropore (mobile) domain (L ³ /L ³)
θ_m	water content in the matrix (immobile) domain (L ³ /L ³)
θ_{sf}	macropore (mobile) domain saturated water content (L ³ /L ³)
θ_A	volumetric water content of spheres (soil peds) (L ³ /L ³)
θ_T	water content of the total macropore-matrix domain (L ³ /L ³)
θ_M	volumetric water content external to spheres (soil peds) (L ³ /L ³)

Appendix B. Analytical solution for batch diffusion experiments

The solution used to model solute diffusion in the batch experiments is adopted from Rao et al. (1980a) who used the analytical solution of Crank (1975) to represent Fickian diffusion from spherical particles in a finite-volume system subject to the following initial and boundary conditions:

$$C_A(a, t) = C_M(t); \quad t \geq 0 \quad (\text{B.1a})$$

$$C_A(r, t) = C_A^0; \quad 0 \leq r \leq a, \quad t = 0 \quad (\text{B.1b})$$

$$C_M(t) = C_M^0 = 0; \quad t = 0. \quad (\text{B.1c})$$

The total mass (M_T) of solute in the aggregate volume (V_A) + water volume (V_M) system is:

$$M_T = V_A \bar{C}_A(t) + V_M C_M(t) \quad (\text{B.2})$$

and the average concentration in a soil aggregate, $\bar{C}_A$, is given as:

$$\bar{C}_A(t) = \frac{3}{a^3} \int_0^a r^2 C_A(r, t) dr \quad (\text{B.3})$$

Then, from the solution solute developed in Crank (1975, pp. 95–96), the value of $\bar{C}_A$ is computed as

$$\bar{C}_A(t) = C^\infty + (C_A^0 - C^\infty) \sum_{n=1}^{\infty} \left\{ \left[\frac{6\beta(\beta+1)}{9+9\beta+q_n^2\beta^2} \right] \exp\left(-\frac{D_a q_n^2 t}{a^2}\right) \right\} \quad (\text{B.4})$$

where the equilibrium concentration C^∞ is:

$$C^\infty = \frac{\theta_A C_A^0 + \theta_M C_M^0}{\theta_A + \theta_M} \quad (\text{B.5})$$

and θ_A and θ_M are the volumetric water contents (ml/cm^3) of the aggregates and external system, respectively, $\beta = \theta_A/\theta_M$, and the values of q_n are the positive non-zero roots of:

$$\tan(q_n) = \left(\frac{3q_n}{3 + \beta q_n^2} \right) \quad (\text{B.6})$$

and can be found for up to $n=6$ in Crank (1975, see Table 6-1). Finally, the concentration external to the spherical particles is:

$$C_M(t) = C_M^0 + \frac{\theta_A}{\theta_M} [C_A^0 - \bar{C}_A(t)] \quad (\text{B.7})$$

All symbols used in Eqs. (B.1a)–(B.7) are defined in Appendix A.

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